$\boxed{\text{expectation}}$

1. $E(X) = \sum_{x : p(x) > 0} x p(x)$

$= \sum_i x_i p(x_i)$ $(x_i \text{ distinct}, \forall i)$

2. $E(g(X)) = \sum_i g(x_i) p(x_i)$

proof: $E(g(X))$

$= \sum_j y_j P\{g(X) = y_j\}$ $(y_j \text{ distinct}, \forall j)$

$= \sum_j y_j \sum_{x : g(x) = y_j} p(x)$

$= \sum_j y_j \sum_{i : g(x_i) = y_j} p(x_i)$

$= \sum_j \sum_{i : g(x_i) = y_j} g(x_i) p(x_i)$

$= \sum_i g(x_i) p(x_i)$ $(x_i \text{ distinct}, \forall i)$

3. $E(g(X)) = \sum_i g(x_i) P(\{r_i\})$

here $X(r_i) = x_i$

proof:

$E(g(X))$

$= \sum_j g(x_j) P(X = x_j)$ (from 2)

$= \sum_j g(x_j) \sum_{r : X(r) = x_j} P(\{r\})$

$= \sum_j \sum_{r : X(r) = x_j} g(X(r)) P(\{r\})$

$= \sum_{r \in \Omega} g(X(r)) P(\{r\})$, ($\Omega$ is sample space)

$= \sum_i g(X(r_i)) P(\{r_i\})$

$= \sum_i g(x_i) P(\{r_i\})$

(r_i distinct, $\forall r$)

(but x_i may be identical)

4. $E(X) = \sum_i x_i P(\{r_i\})$

here $x_i = X(r_i)$

proof: let $g(X) = X$ in 3

$\therefore E(X)$

$= \sum_j x_j P(X = x_j)$

$= \sum_i x_i P(\{r_i\})$ $\blacksquare$

5. $\sum_r p_r^3 \geq \left(\sum_r p_r^2\right)^2$, here $\sum_r p_r = 1$

proof: let X be rv equal to p_r with probability p_r, i.e., $X(r) = p_r$

$P(\{r\}) = p_r$

$\therefore E(X) = \sum_r X(r) P(\{r\}) = \sum_r p_r^2$

$E(X^2) = \sum_r X^2(r) P(\{r\}) = \sum_r p_r^3$

$\because E(X^2) \geq (E(X))^2$

$\therefore \sum_r p_r^3 \geq \left(\sum_r p_r^2\right)^2$

$\blacksquare$